# Fund Risk Assessment Report

Input file: data/sample_nav_data.xlsx

Generated date: 2026-04-09

| Metric | Value |
|---|---|
| Data Frequency | weekly |
| Annualization Factor | 52 |
| Rolling Windows | 12, 26, 52 |
| Inception Annualized Return | 20.98% |
| Inception Return | 84.65% |
| Annualized Volatility | 31.11% |
| Maximum Drawdown | -41.73% |
| Maximum Drawdown Date | 2024-10-18 |
| Sharpe Ratio | 0.6102 |
| Sortino Ratio | 1.0310 |
| Calmar Ratio | 0.5029 |

# Data and Frequency Assessment

The report is based on 166 NAV observations from 2022-11-11 to 2026-01-30. The detected data frequency is weekly, so the analysis uses an annualization factor of 52.

Rolling metrics are calculated with frequency-aware windows: 12, 26, 52. This prevents daily assumptions from being applied to weekly or monthly data.

# Return Performance Commentary

Since inception, the fund generated a total return of 84.65%, equivalent to an annualized return of 20.98% over the observed period.

The year-to-date return is -5.36%, while the trailing one-year return is 14.79%. The win rate is 50.30%, which measures the share of positive periodic return observations.

The return distribution skewness is 0.5154. Positive skewness suggests relatively more upside tail observations, while negative skewness would indicate more downside tail behavior.

# Risk and Drawdown Commentary

Annualized volatility is 31.11%, which is classified as elevated based on the rule-of-thumb thresholds used in this report.

The maximum drawdown is -41.73%, occurring on 2024-10-18. This drawdown is classified as material and has not recovered within the available sample.

Drawdown analysis should be read together with the NAV chart and drawdown frequency chart, because maximum drawdown captures only the deepest loss while frequency captures how persistently the fund stayed below prior highs.

# Tail Risk Commentary

The 95% 1-day equivalent VaR (scaled from weekly returns) is 2.65%, and the corresponding 10-day VaR is 8.37%. These figures estimate downside loss under historical simulation assumptions.

The 99% 1-day equivalent VaR (scaled from weekly returns) is 2.65%, and the corresponding 10-day VaR is 8.37%. The 99% confidence level focuses on more severe tail observations.

The 95% 1-day equivalent CVaR/ES (scaled from weekly returns) is 2.65%, while the 99% 1-day equivalent CVaR/ES (scaled from weekly returns) is 2.65%. CVaR/ES estimates the average loss within the tail, so it is useful for understanding loss severity beyond the VaR threshold.

Note: For non-daily data, short-horizon VaR and CVaR are scaled estimates based on the underlying return frequency.

# Risk-Adjusted Return Commentary

The Sharpe ratio is 0.6102, which is classified as acceptable. The Sortino ratio is 1.0310, which focuses on downside volatility rather than total volatility.

The Calmar ratio is 0.5029, linking annualized return to maximum drawdown. A lower Calmar ratio suggests that return generation has come with a larger path-dependent downside burden.

The Treynor ratio is NaN and the Treynor-Black ratio is NaN. These values are NaN when the input data does not include a benchmark_return column.

# Metrics Summary

| Category | Metric | Value |
|---|---|---|
| Data Profile | input_file | data/sample_nav_data.xlsx |
| Data Profile | start_date | 2022-11-11 |
| Data Profile | end_date | 2026-01-30 |
| Data Profile | observations | 166.0000 |
| Data Profile | data_frequency | weekly |
| Data Profile | annualization_factor | 52 |
| Data Profile | rolling_windows | 12, 26, 52 |
| Return Metrics | inception_annualized_return | 20.98% |
| Return Metrics | inception_return | 84.65% |
| Return Metrics | year_to_date_return | -5.36% |
| Return Metrics | one_year_return | 14.79% |
| Return Metrics | win_rate | 50.30% |
| Return Metrics | return_skewness | 0.5154 |
| Return Metrics | current_consecutive_up_months | 0 |
| Return Metrics | current_consecutive_down_months | 1 |
| Return Metrics | max_consecutive_up_months | 6 |
| Return Metrics | max_consecutive_down_months | 3 |
| Risk Metrics | annualized_return | 20.98% |
| Risk Metrics | annualized_volatility | 31.11% |
| Risk Metrics | max_drawdown | -41.73% |
| Risk Metrics | max_drawdown_date | 2024-10-18 |
| Risk Metrics | max_drawdown_recovery_days | NaN |
| Tail Risk Metrics | var_95_1d_scaled | 2.65% |
| Tail Risk Metrics | var_95_10d | 8.37% |

# Metrics Summary

| Category | Metric | Value |
|---|---|---|
| Tail Risk Metrics | var_99_1d_scaled | 2.65% |
| Tail Risk Metrics | var_99_10d | 8.37% |
| Tail Risk Metrics | cvar_es_95_1d_scaled | 2.65% |
| Tail Risk Metrics | cvar_es_95_10d | 8.37% |
| Tail Risk Metrics | cvar_es_99_1d_scaled | 2.65% |
| Tail Risk Metrics | cvar_es_99_10d | 8.37% |
| Risk-Adjusted Return Metrics | sharpe_ratio | 0.6102 |
| Risk-Adjusted Return Metrics | sortino_ratio | 1.0310 |
| Risk-Adjusted Return Metrics | calmar_ratio | 0.5029 |
| Risk-Adjusted Return Metrics | treynor_ratio | NaN |
| Risk-Adjusted Return Metrics | treynor_black_ratio | NaN |
| Risk Signals | vol_signal | high |
| Risk Signals | drawdown_signal | deep |
| Risk Signals | sharpe_signal | average |
| Risk Signals | tail_signal | low |
| Risk Signals | trend_signal | stable |
| Risk Signals | overall_risk | High |
| Risk Signals | key_risk_driver | Drawdown |
| Risk Signals | monitoring_flag | Escalate |
| Risk Narrative | risk_narrative | The fund exhibits elevated volatility, with sig |

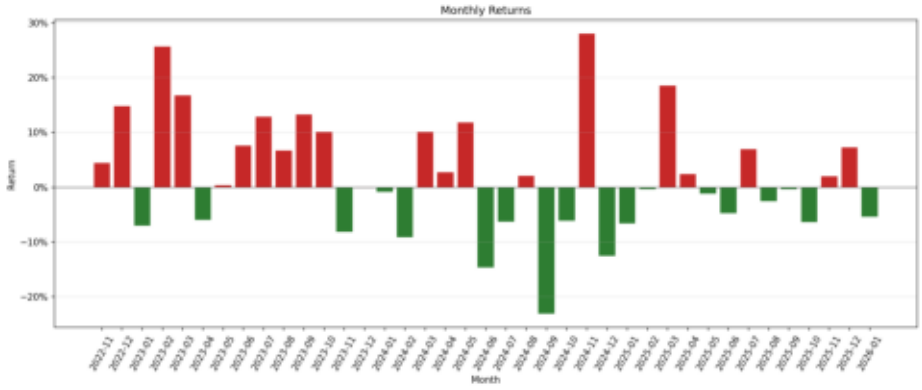
# Risk Commentary

The fund exhibits elevated volatility, with significant drawdown pressure. The fund shows moderate risk-adjusted performance, while risk conditions remain stable. Overall, the portfolio is assessed as High risk, primarily driven by drawdown pressure. Monitoring status: Escalate.

# Return Analysis

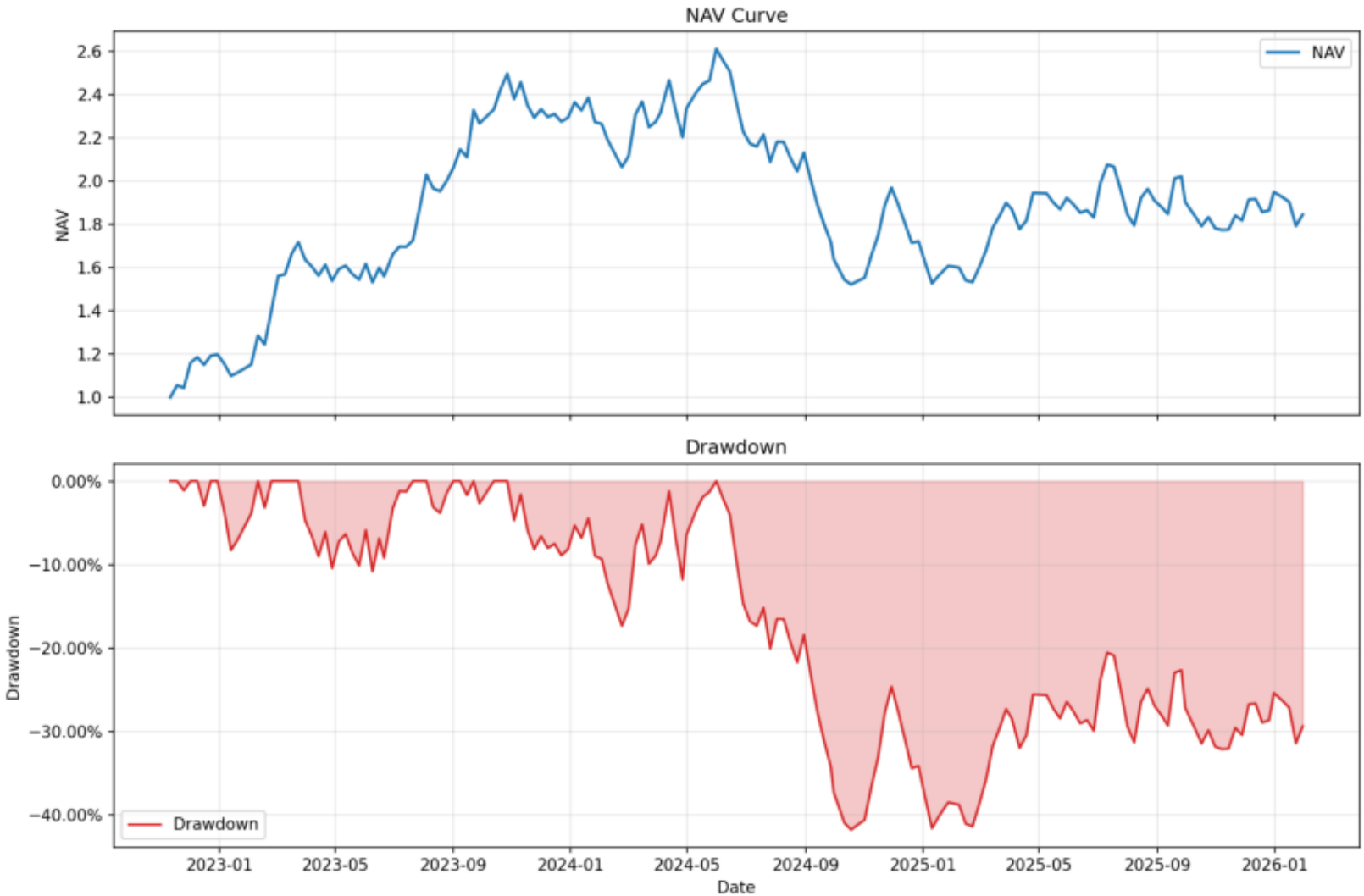
## Annual Return

## Monthly Return

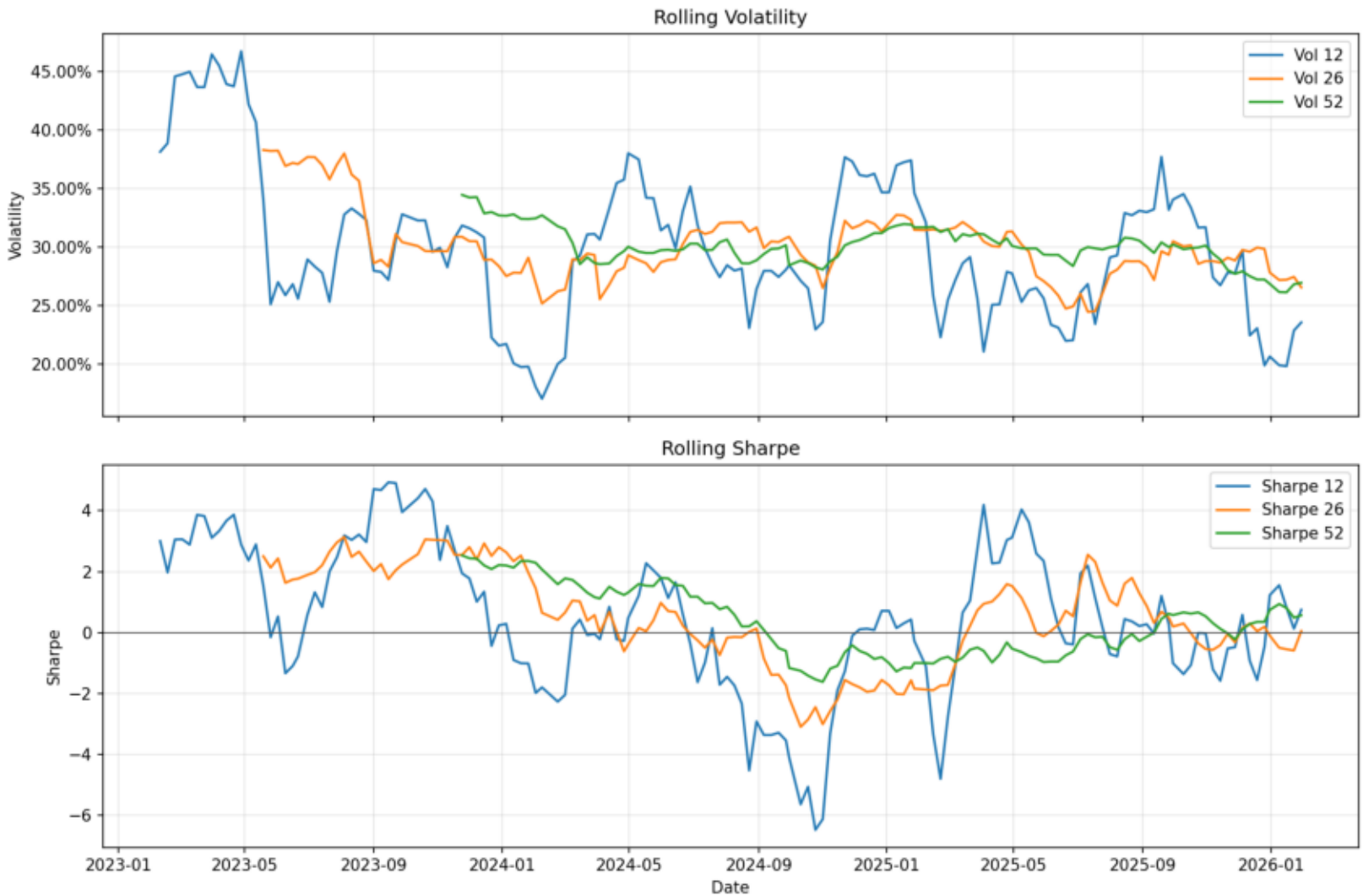



| Year | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | Annual Retur | Win Rate |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2022** | | | | | | | | | | | 4.40% | 14.84% | **19.89%** | **100.00%** |
| **2023** | -6.99% | 25.76% | 16.79% | -5.99% | 0.33% | 7.62% | 12.88% | 6.69% | 13.29% | 10.15% | -8.17% | 0.01% | **91.40%** | **75.00%** |
| **2024** | -0.88% | -9.17% | 10.13% | 2.76% | 11.79% | -14.68% | -6.29% | 2.05% | -23.10% | -6.16% | 28.05% | -12.59% | **-24.94%** | **41.67%** |
| **2025** | -6.61% | -0.38% | 18.61% | 2.37% | -1.14% | -4.72% | 6.96% | -2.53% | -0.33% | -6.36% | 2.02% | 7.24% | **13.28%** | **41.67%** |
| **2026** | -5.36% | | | | | | | | | | | | **-5.36%** | **0.00%** |

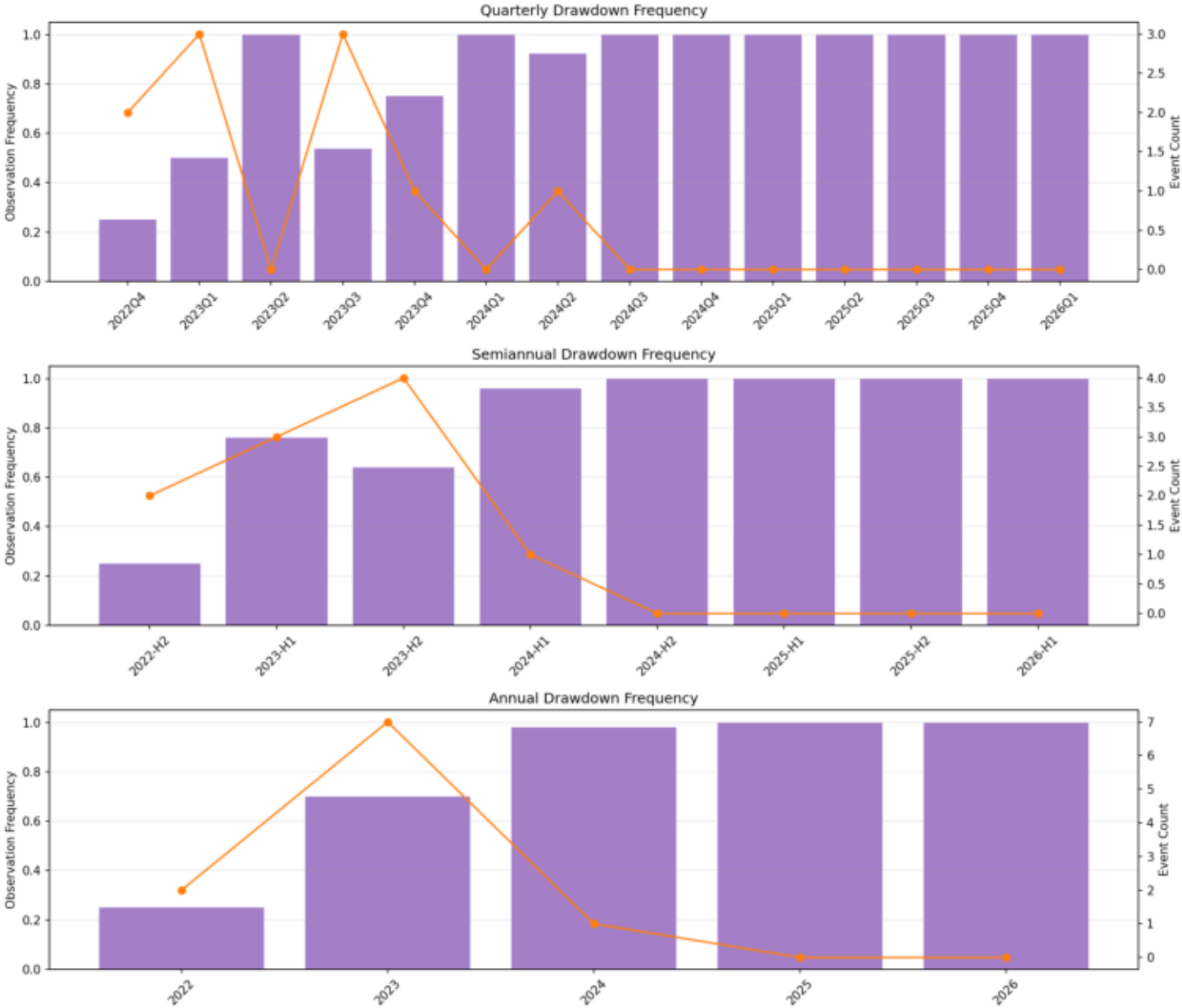
# NAV and Drawdown



# Rolling Risk Metrics



# Drawdown Frequency



# Conclusion and Risk Notes

Overall, this fund demonstrates a historical risk-return profile that should be interpreted through both absolute performance and path-dependent downside behavior.

From a historical perspective, the fund's risk-adjusted profile appears less robust, and further investigation into downside drivers, market sensitivity, and recovery dynamics would be advisable.

Annualized volatility is 31.11%, so ongoing monitoring should focus on whether future realized volatility remains consistent with the historical pattern.

This report is based solely on historical NAV behavior and should be supplemented with strategy-level due diligence, exposure decomposition, and qualitative review before forming any investment decision.